

Manual

SMA-FMO Production Monitoring SMA-FMO 8.2

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1 SMA-FMO Production Monitoring

Purpose

The function package SMA-FMO Production Monitoring includes the following components:

- Workplaces / Machines

Tabular workplace/machine overview showing the current machine status (status, time since when the status has been active). Display of basic information about the workplace/machine.

- Machine information

Detailed display of the selected machine, to be activated from the Workplaces / Machines application .

- TOP 5 shift malfunctions
 - Operations logged on
 - Last 10 malfunctions
 - Operation overview

In addition, the application provides posting functions for the machine, e.g. change status, log on operation, etc. if the appropriate package (SMA-AMF) is available.

- Order overview

Tabular order overview showing orders included in the system. Display of order (header)-specific information such as the article, target quantity, status. Filtering options, e.g. according to status, article

- Operation overview

The app provides greater transparency in production and shows detailed information about the individual operations, if they were selected from the list by the user.

2 Workplaces/Machines

2.1 General

App name	Workplaces/Machines
Short name of app	Workplaces
Function authorization	sma.wpov

The tabular overview *Workplaces/Machines* visualizes the current machine status (status, point in time since when the status is available) including basic information on the workplace/machine. The application helps to increase transparency in the shop floor. The user gets a clear overview of the current statuses of machines and workplaces.

Selection criteria

The application provides the following selection criteria:

Resource

Enter the resource number (workplace number) to select from the workplaces.

Status text

Select the relevant status text to select the status that is displayed.

Field descriptions – List

Image

Shows the picture of the machine stored in the configuration of workplaces and resources.

Note: The picture is loaded when the selection is made. Before selection, a dummy image is shown for the different machines.

Workplace number - workplace designation

Shows the workplace number and workplace name. The system only shows the workplaces that are included in the responsibility area the user is authorized for.

Group

Workplace group of the machine

Status

The respective workplace statuses are presented with the following colors used for the available resource performance accounts:

- RPA 1: dark-green

- RPA 2: red
- RPA 3: pink (fuchsia)
- RPA 4: purple
- RPA 5: black
- RPA 6: dark-gray
- RPA 7: light turquoise
- RPA 8: pale blue
- RPA 9: dark blue
- RPA 10: brown
- RPA 11: light green
- RPA 12: turquoise

Field descriptions – Detail

Image

See description "List".

Workplace number - workplace designation

See description "List".

Status

See description "List".

Status text

Defined status text of the current status

Status since

Date when the status was set

Duration so far

Duration of the status (until now)

Expected end

If specified, the estimated end of the current status

Article

Article logged on to the machine

Article name

Name of the article/item logged on to the machine

MES order number

MES order number of the operation currently logged on. If several operations are logged on to the workplace at the same time, the operation is displayed that was logged on last.

If you click the MES order number, you are redirected to the *Operation overview*.

OP name

Name of the OP logged on

Company

Company of the workplace

Cost center

Cost center of the workplace.

Group

Workplace group of the workplace

2.2 Functions

Subject to the purchased licenses and the available authorizations, the application *Workplaces/Machines* provides the following functions:

**TOP 5 of malfunctions during shift**

Shows the ranking list of the TOP 5 downtimes at the machine including information on the duration and how frequently the malfunction occurred at the respective machine. It is also possible to show the TOP 5 based on the frequency or total duration.

**Running OPs (Operations logged on)**

The list shows all operations logged on to the machine. If you click the operation, you can change to the *Operation overview*.

**Last 10 malfunctions**

The list shows the 10 malfunctions that last occurred at the machine.

**Change status**

Use this function to change the status of the selected machine. A list of the possible machine statuses is provided. Select the relevant status.

Note: This function is included in the package SMA-AMF. Function authorization: sma.setstatus



Operation overview

Click this button to switch to the *Operation overview* and to display detail information on the operation, which was displayed in the detail view of the machine.



Log operation on

Use the dialog *Log operation on* to log an operation on to the currently selected workplace. Select the operation (MES order number) and machine status via search dialog from a list.

Note: All HYDRA standard validation checks are performed.

Note: This function is included in the package SMA-AMF. Function authorization: sma.logon



Log operation off

Use the dialog *Log operation off* to log an operation off from the currently selected workplace. You can use the logoff dialog to enter the operation number (MES order number), yield and scrap quantities, the relevant scrap reason and a new machine status. You can use a search dialog to select the MES order number, the scrap reason and the machine status from a list.

Note: All HYDRA standard validation checks are performed.

Note: This function is included in the package SMA-AMF. Function authorization: sma.logoff



Interrupt operation

Use the dialog *Interrupt operation* to interrupt an operation logged on to the currently selected workplace.

You can enter the operation number (MES order number), yield and scrap quantities, the relevant scrap reason and a new machine status in the dialog. You can use a search dialog to select the MES order number, the scrap reason and the machine status from a list.

Note: All HYDRA standard validation checks are performed.

Note: This function is included in the package SMA-AMF. Function authorization: sma.interrupt



Posting of part quantity (partial confirmation)

You can use the *Partial confirmation* dialog to post part quantities for an operation logged on to the currently selected workplace.

You can enter the operation number (MES order number), yield and scrap quantities and the relevant scrap reason in the dialog. You can use a search dialog to select the MES order number and the scrap reason.

Note: All HYDRA standard validation checks are performed.

Note: This function is included in the package SMA-AMF. Function authorization: sma.partial conf



Change partitioning

Use the *Change partitioning* dialog to change the current partitioning for the workplace currently selected.

The operation number (MES order number) and the new partitioning can be entered in the dialog. You can use a search dialog to select the MES order number from the list of operations logged on.

Note: All HYDRA standard validation checks are performed.

Note: This function is included in the package SMA-AMF. Function authorization: sma.partition



Application settings

In this dialog, you can make specific settings for this application.

- *Reload machine status:*
 - No: The open application is not automatically updated.
 - Cyclic loading: If this option is set, the status of the displayed workplaces/machines is cyclically updated. The duration between two updates is defined via the option *Reload time*.
 - EMQTT: The machine status is promptly updated when the status of the machine changes (via EMQTT). Requirement: You use the centralized MDE and the relevant settings are made. If data collection is not performed via centralized MDE for the machine and if the machine is then not updated via MQTT, an update in the application *Workplaces/Machines* is also not possible.
- *Reload time:* The reload time specifies the time between the status updates. You can define a value between 30 and 300 seconds as reload time.



To enable the settings, click **Save** after having performed the changes and close the application using .

3 Order Overview

3.1 General

App name	Order overview
Short name of app	order
Function authorization	sma.rov

The tabular *Order overview* increases transparency in production and shows the orders processed in the system. The overview provides detailed information for the order header selected. When you click the order, you can directly go to the *Operation overview* and get detailed information on operations.

When you call the *Operation overview* from the *Order overview*, only the operations of the selected order are displayed. Only running and planned orders are shown, finished orders are not displayed.

Selection criteria

The application provides the following selection criteria:

Order status

Select the order status to narrow down the orders displayed. If you do not select any statuses, only the orders in status *Prepared* are displayed. Change this selection by filtering. When a filtering is made using one of the available fields, the default selection by status *Prepared* is discarded.

Order

Order header number

Final article

Article number of the order header

Basic end date

Latest end of the complete order. By default, only the orders are displayed with a latest end before the current day + 5 days in the future.

Field descriptions – List

Order

Order header and article/item number are displayed together

Article name

Article designation of the header article

Status

Shows the status of the order header

Field descriptions – Detail**Order number**

Order header number

If you select the order number, the *Operation overview* opens and the operations of the order are displayed.

Note: You require the function authorization sma.opov to display the *Operation overview*.

Order status

Shows the current order status including status text

Status since

Date when the status was set

Final article

Article of the order

Article name

Article name of the article.

Sales order

Sales order number of the order header

Customer name/designation

Name of the customer stored in the order header

Dates

Scheduled start, scheduled end, basic start date and basic end date of the complete order including all operations

Quantities

Target quantities and recorded of the order in the "basic" quantity unit

4 Operation overview

4.1 General

App name	Operation overview
Short name of app	Operations
Function authorization	sma.opov

The tabular *Operation overview* increases transparency in production and shows detailed information on the operations selected in the list. This application provides a flexible overview of production and the pool of orders.

When you call the *Operation overview* from the *Workplace overview*, only those operations are displayed that are logged on to the machine selected in the *Workplace overview*.

When you call the *Operation overview* from the *Order overview*, only the operations of the order are displayed that was selected in the *Order overview*.

Selection criteria

The application provides the following selection criteria:

Order

Order header number of the operation

MES order number

Combined order and operation number

Final article

Article/item number of the operation

Operation status

Selection by operation status

Field descriptions – List

MES order number

Shows the combined order and operation number.

OP name

Designation from the OP master data

Status

Shows the status of the operation

Field descriptions – Detail

Order number

Order header number of operation

If you select the order number, the order overview opens and the information on the order is displayed.

Note: To display the *Order overview*, you require the function authorization `sma.oro` and the package with the order overview (SMA-FMO) must be available.

MES order number

Shows the combined order and operation number.

Status text

Current status text of the operation.

Status since

Date when the status was set

OP name

Designation from the OP master data

Planned workplace

Workplace where the OP is logged on or where the OP is planned. If you click the workplace, the application *Workplaces/machines* opens and the information on the selected workplace is displayed.

Note: To display the application *Workplaces/Machines*, you require the function authorization `sma.wop` and the relevant package (SMA-FMO) must be available.

Article

Operation article

Article name

Article name of the operation

Dates

Scheduled start, scheduled end, earliest start and latest end of the operation

Quantities

Target quantities and quantities collected for the operation in primary quantity unit

Target setup time

Defined target setup time of the operation

Target processing time

Defined target processing time of the operation

Target remaining run time

Defined target remaining run time of the operation or remaining run time calculated using the formula stored.

Occupancy time

Total of occupancy time

Actual execution time

Total of execution times

Downtime

Total downtime